Supplementary Material S1. Full-Text Screening Results of the Records Assessed for Eligibility

This table lists all 212 unique records retrieved from Scopus and Web of Science (WoS) after language/document-type filtering and duplicate removal, together with the extracted study characteristics and the screening decision. Records are numbered as they appeared in the database search results.

Included studies are shown in bold; within the Model(s) column, the best-performing model is shown in bold. Records whose full texts could not be retrieved are shown in red.

A. Records retrieved from Scopus (n = 203) (including records also indexed in WoS)						
Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
Include	**1.Ali et al. (2027)**	Agriculture / Agri-food systems / Carbon management	Total CO_2-emission prediction and emission-reduction scenario analysis	**RF**	LIME (Local), SHAP (Global), DiCE (Counterfactual)	R^2, MSE, RMSE, MAE, MAPE, MedAE, Max Error, EVS
Include	**2.Özüpak and Aslan (2027)**	Climate / Country-level greenhouse gas emissions	Long-term CO_2 and N_2O emissions forecasting	DT, KNN, AdaBoost, SVM, **RF**	SHAP (Global + Local), LIME (Local)	R^2, RMSE, MAE, MSE, Max Error
Exclude	3.Han et al. (2026)	Integrated energy systems / Low-carbon energy scheduling	MPC operating-cost/tail-risk surrogate prediction and risk-adaptive scheduling	XGBoost, LightGBM, DT, RF, GBDT; MPC/CVaR framework	SHAP, TreeSHAP	MAE, RMSE, R^2, residual standard deviation, Wilcoxon signed-rank test
Include	**4. Ye et al. (2026)**	Transportation / Urban agglomerations / Multimodal transport	Urban transportation carbon-emission (TCE) prediction and driver analysis	LR, RF, XGBoost, LightGBM, **CatBoost**	SHAP (Global + Local)	RMSE, MAE, MAPE, R^2
Include	**5. Sireesha et al. (2026)**	Energy & Industrial sectors / Global GHG emissions	Multi-gas GHG emission prediction and forecasting (CO_2, CH_4, N_2O, F-gases)	**GB**, RF, **XGBoost**, CatBoost, AdaBoost	SHAP (Global + Local), LIME (Local), ICE, PDP	R^2, RMSE, MAE, VAF
Exclude	6.Khorshidi et al. (2026)	Sustainable concrete / Fly ash concrete design	28-day compressive-strength prediction and multi-objective mix optimization	TabNet, SVR, NGBoost, SAINT	SHAP	R^2, RMSE, MAE, MAPE, PBIAS, PCC, SD, cRMSD
Exclude	7.Yan (2026)	Logistics / Transportation	Energy Consumption Forecasting (Regression)	LR, DT, RF, XGB, MLP, **FT-Transformer**	SHAP (Global), LIME (Local)	MAE, RMSE, R^2, MAPE, MSRE, RMSRE, MARE, RMSPE
Include	**8. Saraswat et al. (2026)**	Transportation systems / Vehicle emissions	Vehicle CO_2 emissions prediction	Multi-Layer Perceptron (MLP), **GAO-MLP (MLGO)**, HLOA-MLP (MLHO), XGBoost (XGB), Random Forest (RFR), SVR	SHAP, CAM, RFE	R^2, RMSE, TIC, MAPE, MARE
Exclude	9.Tang (2026)	Energy systems / Global energy market analysis	Energy Price Index and Renewable Energy Share forecasting (Regression)	Extra Trees Regression (ETR), Histogram Gradient Boosting Regression (HGBR), Decision Tree Regression (DTR), optimized using Coyote Optimization Algorithm (COA) and Osprey	SHAP (Global feature importance), CAM – Cosine Amplitude Method (Global sensitivity analysis)	RMSE, R^2, MAE, Scatter Index (SI), Prediction İnterval (PI)

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Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
				Optimization Algorithm (OOA)		
Exclude	10.Keshireddy et al. (2026)	Energy / Oil & Gas Industry	Combustion Performance Forecasting (Efficiency, CO Emissions, Fuel Consumption Index)	Linear Regression (LR), Random Forest (RF), GRU-only RNN, Hybrid Neural Network (Dense + GRU)	SHAP (Global Feature Importance), Permutation Feature Importance	RMSE, MAE, MAPE, R^2
Exclude	11.Elmousalami and Alotaibi (2026)	Construction project management / Earned Value Management	Project budget, cost variance, duration, duration variance and project-risk prediction	ANN, Polynomial Regression, MLR, SVM, DT, RF, AdaBoost, Bagged DT, Gradient Boosting, LightGBM, Extra Trees, XGBoost	SHAP	MAPE, MSE, R^2, Adjusted R^2; Accuracy, Precision, Recall, F1, F2, F-β, ROC-AUC
Exclude	12.Dewangan et al. (2026)	Civil engineering / Sustainable concrete design	Concrete property prediction (slump, flow, compressive strength)	RF, XGB, SVR, DER, PRNN, stacked ensemble (LightGBM meta-learner)	SHAP (global+local), Sobol (global)	RMSE, MAE, R^2, NLL, EC
Exclude	13.Masmali et al. (2026)	Sustainable e-commerce / AIoT / ESG auditing	Sustainability scoring, anomaly/ESG violation detection, audit decisions; carbon-intensity forecasting as a separate green-intelligence component	SVM, RF, LR, CNN, RNN/LSTM, Hybrid Ensemble	SHAP, LIME	Accuracy, Precision, Recall, F1-score, ROC-AUC, Specificity
Exclude	14.Ji et al. (2026)	Residential buildings / Energy retrofit / Climate-responsive building performance	Heating and cooling energy use intensity (EUI) prediction across climate zones	LR, Ridge, Lasso, ElasticNet, RF, GB, XGBoost, LightGBM, NN	SHAP	R^2, RMSE, MAE
Exclude	15.Datta et al. (2026)	Sustainable concrete / Self-healing recycled aggregate concrete	Compressive-strength prediction	Gradient Boosting Regressor, RF, XGBoost, LightGBM	SHAP, PDP	R^2, RMSE, MAE, MAPE
Include	**16. Zhu et al. (2026)**	Urban carbon emissions / Nighttime-light-based estimation / Regional mitigation	City-level carbon emission prediction and heterogeneous driver analysis in the Yangtze River Delta	DT, RF, SVM, KNN, **XGBoost**	SHAP (Global + Local), PDP (Global + Interaction)	R^2, MSE, MAE
Include	**17.Chen et al. (2026)**	Urban sustainability / Carbon emissions	Urban carbon emissions prediction	SVR, RF, LightGBM, AdaBoost, CatBoost, **XGBoost**	SHAP (Global + Local)	R^2, MSE, RMSE, MAE
Exclude	18.Duan et al. (2026)	Composting / Odor control / Waste management	NH_3 and H_2S emissions prediction	XGBoost, RF, LightGBM, SVR, Elastic Net	SHAP	R^2, RMSE, original-scale diagnostics
Exclude	19.Li et al. (2026)	Provincial energy consumption / China / Sustainable energy policy	Provincial total energy consumption prediction and low-carbon policy analysis	RF, GBDT, AdaBoost, XGBoost, LightGBM, CatBoost	SHAP, PDP	R^2, RMSE, MAE, VAF
Exclude	20.Khare and Bhatia (2026)	Sustainable building governance / Building-sector decarbonization	Sustainable building governance and net-zero readiness forecasting	**Agentic AI**, XGBoost, Random Forest, LSTM, ARIMA	Feature Importance Analysis, Local Governance Impact Analysis, Scenario-Based Explanation	Accuracy, RMSE, MAE, R^2

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Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
Exclude	21.Wang et al. (2026)	Green mining / Cemented tailings backfill	Unconfined compressive strength (UCS) prediction	GB, LSTM-XGBoost, MLP-ANN, RBF-SVR, RF, XGBoost-MLP	SHAP	R^2, MSE, RMSE, MAE, MAPE
Exclude	22.Zhang et al. (2026)	Greenhouse agriculture / Plant physiology	Leaf net photosynthetic-rate prediction	XGBoost, ExtraTrees, **XGB–ET Global Blend**, BPNN, SVR, LSTM	SHAP, ALE	R^2, RMSE
Include	**23.Song et al. (2026)**	Agriculture / Dairy cattle slurry storage	Daily CH_4 emission/flux prediction	LR, RF, **XGB**, LGBM, SVR, MLP	SHAP (Global + Local)	R^2, RMSE, MAE
Include	**24.Li et al. (2026)**	Carbon market / Emissions trading / China national ETS	National Carbon Emission Allowance (CEA) price forecasting	Random Walk, ARIMA, Ridge, Lasso, Elastic Net, KNN, SVR, RF, GBDT, **MLP**, Ensemble	SHAP (Global + Local), PDP, Permutation Importance	RMSE, MAE, MAPE, Directional Accuracy, DM-HLN, Clark–West
Include	**25.Ma et al. (2026)**	Carbon market / Emissions trading / Hubei carbon market	Hubei carbon price forecasting and multiscale driver analysis	GRU, LSTM, ELM, CEEMDAN-GRU, CEEMDAN-LSTM, CEEMDAN-ELM, **CEEMDAN-SE-GRU**, CEEMDAN-SE-LSTM, CEEMDAN-SE-ELM, VC Ensemble	SHAP (Global + Local / multiscale), TVP-SV-VAR	MSE, MAE, RMSE, MAPE, R^2
Exclude	26.Gallego Fernández et al. (2026)	Cement industry / Biomass co-processing	Fuel-flow prediction and indirect reconstruction of combustion CO_2 emissions	RF	SHAP	R^2, MAE, RMSE
Include	**27.Shaik et al. (2026)**	Transportation systems / Vehicle emissions	Vehicular CO_2 emission prediction (g/km)	TabNet optimized with HEO and **FFO**	TabNet Feature Masks (Global + Local), SHAP (Global + Local), LIME (Local)	R^2, RMSE, MAE, MAPE
Include	**28.Chen et al. (2026)**	Transportation systems / Transportation carbon emissions	Transportation CO_2 emissions prediction and multi-scenario forecasting	Extra Trees (**ET**)	SHAP (Global + Local)	MSE, RMSE, R^2
Include	**29.Song et al. (2026)**	Power systems / Substations / Carbon accounting	Dynamic substation carbon-emission-factor (CEF) prediction	RF, GBM, LSTM, Pure Transformer, **MOIRAI-2 + UPINN**	Any-Variate Attention (Global + Local), UPINN Physical Decomposition (Global + Local), Counterfactual Reasoning (Local)	MAE, RMSE, R^2
Exclude	30.Paul et al. (2026)	Agriculture / Dairy manure / Anaerobic digestion	Methane yield and methane emissions prediction	Support Vector Regression (SVR), Artificial Neural Network (ANN), Extreme Gradient Boosting (XGBoost), Random Forest (RF)	SHAP	R^2, MAE, RMSE
Exclude	31.Ban (2026)	Sustainable resource allocation / Green economy	Sustainability score prediction and resource allocation optimization	Attention-LSTM, NSGA-II	SHAP	RMSE, MAE, MSE, MAPE, R^2
Include	**32.Wani et al. (2026)**	Construction materials / Sustainable concrete	Cradle-to-gate CO_2 emissions prediction in GGBFS-based geopolymer concrete	XGBoost (XGB), LightGBM (LGBM), **CatBoost (CB)**, AdaBoost (AB), Artificial Neural Network (ANN)	SHAP (Global + Local), LIME (Local), PDP	R^2, MAE, RMSE, MSE, NSE, MAPE

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Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
Include	**33.Luo et al. (2026)**	Transportation / Tunnel emissions / Road slope analysis	Instantaneous vehicle CO_2 emissions prediction in tunnel sections	Exponential VSP model, MLR, RF, XGBoost	R^2, RMSE	SHAP (Global), ICE (Local)
Exclude	34.Yan et al. (2026)	Copper smelting / Coal blending optimization	Coal blending ratio prediction	SVR, RF, CatBoost, XGBoost, LightGBM, ExtraTrees	SHAP, PDP, ICE	R^2, MSE, RMSE, MAE, VAF
Exclude	35.Nageshwaran et al. (2026)	Sustainable construction / One-part geopolymer binders	28-day compressive-strength prediction	XGBoost, Random Forest, SVR, Linear Regression	SHAP, TreeSHAP	R^2, RMSE, MAE
Exclude	36.Das et al. (2026)	Solar thermal systems / Parabolic trough collectors	Outlet temperature prediction of PTC	M5P model tree, Elastic.Net regression, CFD-supported framework	M5P, Elastic.Net	R^2, MAE, RMSE
Exclude	37.Verma et al. (2026)	Animal agriculture / Environmental compliance	Environmental compliance-status classification	LR, RF, XGBoost, DNN, EBM	EBM intrinsic interpretability, SHAP, LIME	Accuracy, Precision, Recall, F1-score, AUC
Exclude	38.Gupta et al. (2026)	Livestock enterprises / Environmental law compliance	Environmental compliance classification and regulatory-risk prediction	Transformer-based NLP, ML classifiers, semantic knowledge graph	XAI-based explainable decision support	Accuracy, Precision, Recall, F1-score, Risk Prediction Accuracy
Exclude	39.Wankhade et al. (2026)	Urban ecology / Environmental management	Review and conceptual AI decision-support framework covering pollution, waste, water and carbon-emission monitoring	-	-	-
Not retrieved	40.Ai and Lu (2026)	Not retrieved	Not retrieved	Not retrieved	Not retrieved	Not retrieved
Include	**41.Yang et al. (2026)**	Carbon market / Emissions trading / China's eight ETS pilots	Multi-scale carbon price forecasting and driver analysis	**XCPFNet (CEEMDAN-CNN-LSTM-Transformer)**, EMD-TI-LSTM, CEEMDAN-LSTM-Transformer, CMD-SVR-LSTM	Improved/Clustered SHAP (Global + Local)	MAE, MAPE, MSE, RMSE, R^2, PIPC, PINAW
Include	**42.Özüpak (2026)**	Aviation / Air transport sustainability	Aviation CO_2 emissions prediction and high-emission detection	Physics-guided baseline, LSTM, 1D-CNN, Transformer, Hybrid model, **Stacked Ensemble**	SHAP (Global + Local), LIME (Local)	RMSE, MAE, R^2, AUC, NSE, Theil U2
Exclude	43.Iftikhar et al. (2026)	Civil engineering / Sustainable construction materials	Flexural strength prediction of cement-free plastic–sand composites	Support Vector Regression (SVR), Extreme Gradient Boosting (XGB), Gene Expression Programming (GEP)	SHAP (global feature importance)	R^2, MAE, RMSE, RMSLE
Include	**44. Sun et al. (2026)**	Carbon market / Emissions trading / Shanghai and Hubei ETS	Carbon price forecasting and driver analysis	**CaStNet**, CEEMDAN-SE-VMD + LSTM-Transformer, ALA-LSTM-Transformer, Informer, DLinear, LSTM-Transformer, TCN, CNN-LSTM, Transformer, GRU,	SHAP (Global), LIME (Local)	RMSE, MAE, MAPE, R^2, DM test

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Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
				XGBoost, SVR, ARIMA		
Include	**45.Chai et al. (2026)**	Enterprise sustainability / Carbon management	Near-real-time Scope 1–2 emissions estimation and reduction recommendation	Transformer, CNN, BiLSTM, Ensemble (Transformer + CNN + BiLSTM); Optimization: NSGA-III, PSO, SA	SHAP (Global + Local + Interaction), LIME (Local)	MAPE, RMSE, Pearson r, Emission reduction (%), ROI, NPV, Payback period
Exclude	46.Li et al. (2026)	Mining / Grouting / Water-inrush control	Unit grouting amount prediction	KNN, SVM, RF, XGBoost, CNN, MLP, BiTCN–BiGRU–Attention	SHAP, ALE	R^2, MAE, RMSE, WMAPE
Not retrieved	47.Zhang et al. (2026)	Not retrieved	Not retrieved	Not retrieved	Not retrieved	Not retrieved
Not retrieved	48.Xu et al. (2026)	Not retrieved	Not retrieved	Not retrieved	Not retrieved	Not retrieved
Exclude	49.Hayawi et al. (2026)	Energy systems / District cooling	Cooling load forecasting	Linear Regression (LR), Random Forest (RF), Gradient Boosting (GB), XGBoost (XGB)	SHAP (Global)	MSE, MAE, RMSE, R^2, Adjusted R^2
Include	**50.Gungor et al. (2026)**	Smart buildings / Campus energy management	Building-level and campus-wide annual CO_2 emissions estimation	**ANN,** SVR, SVR_BO, GPR, LSBoost, LSBoost_PI, Bagged Trees, Ridge, Tree, Stacking Ensemble	SHAP (Global + Local), Permutation Importance (Global), Ablation Analysis (Global)	RMSE, MAE, MSE, Bias, R^2, MAPE, NRMSE, EVS, Corr, RMSLE
Include	**51.Seo et al. (2026)**	Construction / Pre-construction stage	Construction equipment carbon emissions forecasting	**Decision Tree (DT)**, Random Forest (RF), XGBoost (XGB), **Support Vector Regression (SVR),** Multi-Layer Perceptron (MLP), Ensemble (**DT+SVR** etc.)	SHAP (Global + Local + Interaction)	MAE, SMAPE
Exclude	52.Atique et al. (2026)	Automotive engineering / Fuel efficiency / Environmental sustainability	Combined MPG prediction and CO_2 estimation from fuel consumption	14 ML algorithms + MLP, Bayesian XGBoost, stacking ensemble	SHAP, LIME, permutation importance	R^2, RMSE, MAE
Exclude	53.Wang et al. (2026)	Agriculture / Carbon emissions	Agricultural carbon emission efficiency (ACEE) forecasting	XGB, **BPNN (NSGA-II optimized)**, GTWR	SHAP (Global)	R^2, AICc, RSS
Include	**54.Shi et al. (2026)**	Energy / Urban sustainability / Carbon emissions	Energy-related CO_2 emission estimation and spatial reconstruction	LR, RFR, SVR, **XGBoost**	SHAP (Global + Interaction)	R^2, RMSE
Exclude	55.Zhang et al. (2026)	Urban planning / Street vitality / Low-carbon urban renewal	Diagnostic analysis of street vitality–low-carbon coordination	XGBoost, StackingRegressor and spatial diagnostic models	SHAP, OPGD, LOWESS	R^2, RMSE, MAE and coordination diagnostics
Exclude	56.Franck-stève et al. (2026)	Energy systems / Short-term load forecasting	Electricity load forecasting	RS-ANFIS, LSTM, GRU, hybrid fuzzy-time-series approaches	Interpretable neuro-fuzzy inference / fuzzy rule analysis	RMSE, MAE, MAPE, MdRAE, R^2
Exclude	57.Mirzaei et al. (2026)	Sustainable concrete / Low-carbon concrete materials	28-day control-relative compressive-strength ratio prediction	RF, SVR, XGBoost, LightGBM, CatBoost	SHAP	R^2, RMSE, MAE
Exclude	58.Reina-Jiménez et al. (2026)	Time series forecasting / Model selection	Forecasting model selection and	Regression Tree, M5P, RF, XGBoost, LSTM, CNN, N-	Decision-tree meta-learner / interpretable rules	Accuracy, computational efficiency, energy

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Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
			performance benchmarking	BEATS, Transformer, iTransformer		consumption, CO_2 emissions
Include	**59.Wang et al. (2026)**	Construction / Carbon emissions	Carbon emission prediction	LR, RF, BPNN, **XGB**	SHAP (global), PDP (feature effects)	R^2, MAE, RMSE, MAPE
Exclude	60.Adamu et al. (2026)	Sustainable construction / Alkali-activated concrete	Compressive strength prediction	CNN, LSTM, Random Forest, XGBoost	SHAP	PCC, R^2, adjusted R^2, WI, MSE, MAE, MAPE, RMSE
Include	**61.Cao et al. (2026)**	Construction materials / Steam-cured concrete	Carbon emission and concrete performance prediction in steam-curing processes	Random Forest (RF), BO-RF, XGBoost, BO-XGBoost, CatBoost, **BO-CatBoost**; improved NSGA-III (optimization)	SHAP (Global &Local)	R^2, RMSE, MAE
Exclude	62.Rawat et al. (2026)	Cybersecurity / Online social network security	Phishing and impersonation detection	XGBoost, LightGBM, CatBoost, Logistic Regression meta-learner	SHAP, LIME	Accuracy, F1-score, AUC, latency, energy, CO2e, SSR
Exclude	63.Katlay and Turk (2026)	Construction materials / Biochar-modified cementitious composites	Compressive strength modeling	RF, ETR, GBM, XGB, stacking ML models	SHAP, ICE	R, R^2, RMSE, MAE, MAPE, RRMSE
Not retrieved	64.Chen et al. (2026)	Not retrieved	Not retrieved	Not retrieved	Not retrieved	Not retrieved
Exclude	65.Selicati et al. (2026)	Residential energy systems / EV charging / PV-BESS	Optimal EV charging schedule prediction	RF, XGBoost, SVR, KNN, FNN/MLP, TabNet	SHAP	R^2, RMSE, CVRMSE, U95, MAE, MAPE
Exclude	66.Lee et al. (2026)	Hydrogen production / PEM water electrolysis	Hydrogen flow rate prediction	**CatBoost**, DT, ET, KNN, LGBM, RF, XGB	SHAP, PDP	R^2, MSE, RMSE, MAE
Exclude	67.Liu et al. (2026)	Wastewater treatment / Environmental engineering	Operational electricity consumption forecasting	LR, SVR, DT, RF, **ET**, GBR, XGBoost, MLP	SHAP, PDP	R^2, MSE, RMSE, MAE, MAPE
Exclude	68.Lotun et al. (2026)	Buildings / Passive thermal measures	Indoor heat gain prediction	MLR, RR, LR, HR, RFR, GBR, XGBR, DNN, CNN, LSTM, GRU, HCL, HCLG, TN	SHAP	R^2, RMSE
Exclude	69.Wang et al. (2026)	Solar energy / solar nowcasting	Review of solar nowcasting methods	-	-	-
Include	**70.Ranjan and Madanayak (2026)**	Transportation / Freight Operations	Carbon emission intensity / CO_2 emissions modeling	MLR, LASSO, BRR, RF, **XGBoost**	SHAP (global & local), Partial Dependence Plots	R^2, RMSE, MAE, MSE
Include	**71.Sireesha and Sheik (2026)**	Agriculture / Global GHG emissions	CO_2, CH_4, and N_2O emissions prediction	CNN, RNN, GRU, DFNN	SHAP (Global + Local), LIME, ICE, PDP	RMSE, MAE, R^2, VAF
Include	**72.Farooq et al. (2026)**	Transportation / Vehicle emissions	Fuel consumption and CO_2 emission prediction	Random Forest, AdaBoost, Gradient Boosting, Extra Trees, XGB Regressor, LGBM Regressor, Linear Regression, Ridge Regression, Lasso Regression, SVR	LIME (Local)	R^2, MAE, MSE, RMSE
Include	**73.Fang et al. (2026)**	Buildings / Residential	Hourly operational carbon emission prediction	DCG-TCN-GCN, AR, DT, SVM, FNN, GPR, Sageformer,	SHAP (Global + Local)	Corr, RSE

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		building operations		TPA-LSTM, MT-GNN		
Include	**74.Dahmani et al. (2026)**	Climate / Environmental forecasting	CO_2 emissions and temperature forecasting	Multi-task Conv1D-LSTM	LIME (Local)	MAE, MSE, RMSE, R^2
Include	**75.Qader and Albalooshi (2026)**	Energy / Power stations	Power-station CO_2 emissions forecasting	XGBoost, Random Forest, LSTM, SVR	SHAP (Global + Local), LIME (Local)	RMSE, MAE, R^2, MAPE
Include	**76.Ghannam (2026)**	Climate / National emissions forecasting	National CO_2 emissions forecasting in Australia	ARIMA, **Random Forest**, XGBoost, SVR, GPR, ELM, ISSA-ELM, MLP, LSTM, RNN, GRU, stacking regressor, enhanced stacking regressor	SHAP (Global), Tree-based feature importance (Global)	MSE, RMSE, R^2, MAE, MAPE, MedAE, MSLE
Exclude	77.Shi et al. (2026)	Fisheries / Marine environment	Yellowfin tuna fishing grounds prediction	RF, Extra Trees, AdaBoost, XGBoost, LightGBM, CatBoost	SHAP, LIME	Accuracy, Precision, Recall, F1-score, AUC
Include	**78.Ramalingam et al. (2026)**	Sustainable energy / Biofuel combustion	CO_2 emissions and engine performance-emission prediction	XGBoost, AdaBoost, **GBM**	SHAP (Global + Local), Feature Importance Analysis (Global)	R^2, RMSE, MSE, MAE
Exclude	79.Marijan et al. (2026)	Maritime transport / Ship fuel consumption	Review of fuel consumption estimation and optimization methods	-	-	-
Not retrieved	80.Luo et al. (2026)	Not retrieved	Not retrieved	Not retrieved	Not retrieved	Not retrieved
Exclude	81.Mohsen and Mohsen (2026)	Agriculture / Livestock supply chains	CH_4 and N_2O emissions prediction from manure management	Random Forest Regression, XGBoost, SVR	-	RMSE, MAE, R^2
Exclude	82.Alatefi et al. (2026)	Carbon capture / Porous liquids	CO_2 uptake prediction in porous liquids	**TabPFN**, GRNN, LGBM	SHAP (Global)	R^2, MSE, RMSE, AARE
Include	**83.He et al. (2026)**	Carbon market / Emissions trading / China and EU ETS	Multi-market carbon price forecasting and risk monitoring	SVR, LSTM, GRU, BiGRU, TCN; CEEMDAN–CNN–SVR, CEEMDAN–CNN–LSTM, CEEMDAN–CNN–GRU, CEEMDAN–CNN–BiGRU, CEEMDAN–CNN–TCN; CEEMDAN–SVR–Attention, CEEMDAN–LSTM–Attention, CEEMDAN–GRU–Attention, CEEMDAN–BiGRU–Attention, CEEMDAN–TCN–Attention; CEEMDAN–CNN–SVR–Attention, CEEMDAN–CNN–LSTM–Attention, CEEMDAN–CNN–GRU–Attention, **CEEMDAN–CNN–BiGRU–Attention**, CEEMDAN–CNN–TCN–Attention	SHAP (Global + Local), LIME (Local)	MSE, RMSE, MAE, MAPE, R^2

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Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
Exclude	84.Ezzat et al. (2026)	Agri-food systems / Environmental impact	Average temperature rise prediction from agri-food CO_2 emissions	Random Forest (RF), Support Vector Regression (SVR), **MSA-optimized DNN**	SHAP (Global + Local)	R^2, MSE, RMSE, MedAE, MAE, LogCosh
Exclude	85.Sungheetha et al. (2026)	Healthcare / Depression treatment optimization	Psilocybin treatment response and optimization modeling	Digital twin + Bayesian optimization + multimodal depression detection framework	XAI framework (multiple interpretability methods)	Accuracy, explainability score, safety index, compliance metrics
Not retrieved	86.Ma et al. (2026)	Not retrieved	Not retrieved	Not retrieved	Not retrieved	Not retrieved
Include	**87.Ma et al. (2026)**	Climate change / Carbon emissions	CO_2 emissions prediction	Random Forest (RF), XGBoost, LightGBM, CatBoost, **Multilayer Perceptron (MLP)**	SHAP (Global + Local)	R^2, MSE, RMSE, MAE, MAPE
Include	**88.Zheng and Shuang (2026)**	Buildings / Construction / Regional carbon analysis	Embodied carbon emissions prediction in China's building sector	SVM, ANN, RF, XGBoost, **CatBoost,** BP, LSTM	SHAP (Global + Local)	MAE, MAPE, RMSE, R^2
Exclude	89.Zhang et al. (2026)	Construction management / Concrete resource allocation	Concrete resource mismatch and daily concrete consumption prediction	XGBoost, RF, LR, MLP, GAM	SHAP	R^2, MAE, MSE, RMSE, MAPE
Include	**90.Saleem et al. (2026)**	Intelligent transportation / CAVs	Fuel consumption and CO_2 emissions prediction	Federated learning framework, FMS-selected global model, Extra Trees (selected global model)	LIME (Local), SHAP (Global)	R^2, RMSE, MAE
Exclude	91.Aljubairi et al. (2026)	Urban transportation / Shared mobility	Energy-intensity prediction (kJ/pax-km) and policy scenario analysis	XGBoost, LightGBM, RF, ExtraTrees, GradientBoosting	SHAP, LIME	R^2, RMSE, MAE, MAPE
Exclude	92.Sheng et al. (2026)	Sustainable cement / Carbonatable binders	Compressive-strength prediction under carbonation curing	ETR, RFR, SVR, XGBoost	SHAP	RMSE, R^2
Exclude	93.Liu et al. (2026)	Sustainable infrastructure / Dredged sediment stabilization	28-day unconfined compressive strength (UCS) prediction	ANN	SHAP, 2D-PDP	R^2, MSE, RMSE, MAE
Include	**94.Ren et al. (2026)**	Urban sustainability / Urban morphology / Carbon management	Sectoral and total urban carbon emissions prediction	ResNet-18, **XGBoost**, mediation–moderation models	Grad-CAM (Spatial / Local), SHAP (Global + Local)	R^2, MAE, RMSE, EVS, classification accuracy/precision
Include	**95.Bhowmik et al. (2026)**	Forest monitoring / Climate mitigation / Land-use change	Tree cover loss and CO_2 emissions forecasting	LR, **RF, XGBoost**, LSTM	SHAP (Global)	R^2, RMSE, MAE
Exclude	96.Guo and Jin (2026)	Biomass conversion / Hydrothermal processing	Formic acid yield prediction	XGBoost, RF, GBDT, AdaBoost, LR, SVR, KNN, Ridge	SHAP, PDP	R^2, RMSE, MAE
Not retrieved	97.Zheng-Yi et al. (2026)	Not retrieved	Not retrieved	Not retrieved	Not retrieved	Not retrieved
Include	**98.Zawaideh (2026)**	High-rise construction / BIM-based risk	Cost overrun, schedule delay, and embodied-carbon risk prediction	**XGBoost**, LightGBM, CatBoost, RF, SVR, ANN	SHAP (Global)	R^2, RMSE, MAE, MAPE

A. Records retrieved from Scopus (n = 203) (including records also indexed in WoS)						
Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
		management / Embodied carbon				
Include	**99.Jiang et al. (2026)**	Land use / Regional carbon management	County-level land-use carbon emission forecasting	**XGBoost,** RF, PLUS, CA–Markov, InVEST, Carbon Emission Coefficient Method	SHAP (Global + Local + Interaction)	R^2, Kappa
Include	**100.Sayghe et al. (2026)**	Power systems / Energy forecasting / Smart grid	Multi-task forecasting of load, price, carbon emissions, and renewable generation	ARIMA, Prophet, XGBoost, LGBM, LSTM, GRU, TCN, TFT, N-BEATS, Informer, Autoformer, PatchTST, TimesFM, Chronos, Moirai, Moirai-MoE, TimesFM-FT, Chronos-FT, Moirai-MoE-FT, **GridFM**	SHAP (Global + Local), Attention Visualization (Global)	MAPE, RMSE, MAE, CRPS
Exclude	101.Dong et al. (2026)	Carbon capture / Biochar / Gas separation	CO_2 adsorption, CO_2/N_2 selectivity, and CO_2/CH_4 selectivity prediction	**XGB**, RF, LightGBM	SHAP (Global), SHAP-based Partial Dependence Analysis (Global)	R^2, RMSE
Not retrieved	102. Guo et al. (2026)	Not retrieved	Not retrieved	Not retrieved	Not retrieved	Not retrieved
Exclude	103.Su et al. (2026)	Sustainable construction / Geopolymer binders / Dredged sludge modification	UCS prediction and mix proportion optimization	PMV-IEL, SVM, ELM, BPNN, GRU, KNN, CNN, NSGA-II	SHAP, knowledge graph	R^2, MAE, RMSE
Exclude	104.Nguyen et al. (2026)	Maritime logistics / Green shipping	Port congestion and relative vessel fuel-demand prediction	RF, XGBoost, LightGBM	SHAP	R^2, RMSE
Exclude	105.Ceken and Tuncal (2026)	Sustainable aviation / AI applications in aviation	Review of AI applications for fuel efficiency, emissions, maintenance, and aviation sustainability	-	-	-
Exclude	106.Nyberg et al. (2026)	Telecommunications / Energy-efficient cellular radio systems	PA voltage-level classification and PRB utilization forecasting	Random Forest, XGBoost, KNN, SVM, FFNN, Hybrid LSTM, Hybrid CNN, TFT, ARIMA	SHAP	Energy savings, underestimations, F1 score, MSE, MAE
Exclude	107.Erlangga and Cho (2025)	Energy management / Household electricity	Power demand prediction	CNN-LSTM with attention, Bayesian network	Causal XAI, Bayesian network, CAM, attention analysis	MSE, RMSE, MAE
Include	**108.Zhou et al. (2025)**	Carbon market / Emissions trading / EU, New Zealand, Hubei and China national ETS	Carbon price point and interval forecasting and price-driver analysis	**DNMADE (MOWAA-AttESN + DHKDE)**, GS-BPnn, GS-ESN, GS-LSTM, GS-CNNLSTM, GS-Transformer, GS-DLinear, MGOA-ORELM, CWOA-ELM, SSA-BPnn, SMA-BiLSTM, HKDE, LUBE, QTCN, GPR	SHAP (Global + Local), AW	MAE, RMSE, MAPE, MSPE, IA, DA, PICP, PINAW, AIS, CWC
Include	**109.Alam et al. (2025)**	Transportation / Vehicle emissions	Vehicle CO_2 emissions prediction	DT, KNN, SVR, XGBoost,	SHAP (Global + Local)	R^2, MSE, RMSE, MAPE

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Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
				AdaBoost, CatBoost, DT + KNN + XGBoost, LSTM, BiLSTM, **CarbonMLP**		
Include	**110.Şahin (2025)**	Maritime transportation / Passenger vessels / Carbon emissions	Hourly CO_2 emissions prediction from passenger ships	Random Forest, **XGBoost**	SHAP (Global)	R^2, RMSE, MAE, MAE%
Exclude	111.Alfayan and Hagras (2025)	Smart grid / Energy management	Future kWh energy consumption prediction	Type-1 FLS, Type-2 FLS, Neural Network	Type-2 fuzzy logic XAI	RMSE, MAE
Exclude	112. Novielli et al. (2025)	Soil science / Climate response / Agriculture	Soil respiration sensitivity (Q10) prediction	Extra Trees Classifier	SHAP, t-SNE, clustering	Accuracy, Precision, AUC-ROC, AUC-PRC
Exclude	113.Faisal et al. (2025)	Urban policy / Carbon trading governance	Evaluation, clustering, and prediction of city-level carbon trading strategies	PSOM, Random Forest, MOO-MCDM	interpretable ML framework	classification accuracy / clustering results
Include	**114.Zhang et al. (2025)**	Campus / Buildings / Community-scale carbon management	CO_2 emissions prediction and optimization for a university campus	**Random Forest Regressor (RF)**	SHAP (Global), Pearson Correlation Analysis (Global)	R^2, RMSE
Include	**115.Liu et al. (2025)**	Civil engineering / Bridge construction / Carbon emissions	Bridge construction CO_2 emission prediction	**GBRT**, XGBoost, CatBoost, RF, AdaBoost, MLP, DT, SVR	SHAP (Global + Local), PDP (Global), ICE (Local)	MSE, MAE, RMSE, R^2, CWI
Include	**116.Huang et al. (2025)**	Urban transportation / Neighborhood-level planning	Neighborhood-level road traffic CO_2 emission prediction	**Extreme Gradient Boosting (XGBoost)**	SHAP (Global + Local), GAM (Nonlinear effects)	R^2, RMSE, MAE
Exclude	117.Shmuel et al. (2025)	Wildfire science / Climate risk	Lightning-ignited wildfire prediction and risk estimation	Logistic Regression, Random Forest, AutoGluon, XGBoost	SHAP	Accuracy, AUC
Exclude	118.Tripathi et al. (2025)	Retail / Sustainable decision-making	Customer retention forecasting	Quantile regression, conformal prediction	SHAP	forecast interval / calibration metrics
Exclude	119.Bhagat et al. (2025)	Sustainable construction / 3D-printed materials	Material ranking, life cycle sustainability assessment, and optimization	Random Forest, GPR, PPO	SHAP	sustainability grade / optimization gains
Exclude	120.Al Mazroa (2025)	Industrial energy management / Smart zones	Multi-horizon load and price forecasting with carbon-aware scheduling	**PatchTST**, LSTM, GRU, GBR, SAC; scheduling, MPC; scheduling baseline, Rule-Based Scheduling (baseline)	SHAP (Global + Local), Attention (Temporal + Local), Attention–SHAP Alignment (Local)	R^2, MAE, RMSE, MAPE, nMAE; CO_2 Reduction (%), Cost Savings (%)
Exclude	121.Venkatachalam et al. (2025)	Biodiesel / Engine optimization / Combustion	BTE, BSFC, EGT, NOx, CO, HC, and smoke prediction with blend/load optimization	SVR, XGBoost, RSM, NSGA-II	SHAP	R^2, RMSE, MAE
Exclude	122.Zhao et al. (2025)	Wildfire / Global fire emissions / Climate change	Global burned-area prediction followed by wildfire carbon-emission projection	Interpretable ML model for burned-area prediction	SHAP	R^2; emission projection comparisons
Exclude	123. Altiary & Hussain (2025)	Carbon market / ETS / Carbon price forecasting	Systematic review of explainability in carbon price forecasting systems	-	-	-

A. Records retrieved from Scopus (n = 203) (including records also indexed in WoS)						
Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
Exclude	124.dos Santos et al. (2025)	Catalysis / CO_2 methanation / Renewable fuels	CO_2 conversion and CH_4 selectivity prediction for Ni-based catalysts	**XGBoost**, LGBM, RF, GBDT	SHAP (Global + Local), PDP (Global + Interaction)	R^2, MAE, MSE, RMSE
Exclude	125.Belkadi et al. (2025)	GHG monitoring / North Africa / Satellite-based methane monitoring	Next-day methane-emission forecasting	LSTM, GRU, CNN-LSTM, CNN-GRU, QLSTM, QGRU, CNN-QLSTM, CNN-QGRU, QCNN-LSTM, QCNN-GRU	-	MAE, MSE, RMSE, R^2
Exclude	126.Lu and Li (2025)	Building energy benchmarking / Energy-efficient buildings	Annual and monthly building energy consumption and site EUI prediction; building-stock clustering	MLR, Random Forest, LGBM, SOM	SHAP, OLS sensitivity analysis, feature importance	R^2, RMSE, acceptance interval %, Calinski–Harabasz score, Silhouette score
Exclude	127.Hussain and Soliman (2025)	Sustainable construction / Biochar-based cementitious composites	Compressive strength, flexural strength, and fracture energy prediction	MAR, SPR, XGBoost; BO, GSO and RSO optimization	SHAP, PDP	R^2, RMSE, MAE, WI, KGE
Exclude	128.Iwakin et al. (2025)	Water–energy microgrids / Distributed energy resource management	Optimal power-dispatch prediction for distributed energy resources	GBRT, MLP, weighted ensemble; PSO optimization	SHAP	R^2, MAE, RMSE, NRMSE
Exclude	129.Du et al. (2025)	Plasma-catalytic ammonia synthesis / Process optimization	NH_3 concentration and energy-efficiency prediction	Ridge Regression, kNN, Decision Tree, Random Forest, MLP; Bayesian optimization and NSGA-II	SHAP	R^2, RMSE, MSE, HV, SP
Include	**130.Bilotta et al. (2025)**	Urban sustainability / Smart cities / Carbon emissions	Fine-scale urban CO_2 emission estimation using open city-service and socioeconomic data	LR, OLS, **XGBoost**, GCN	SHAP (Global + Local)	RMSE, MAE, MSE, MAPE, R^2
Include	**131.Yuan et al. (2025)**	Transportation / Vehicle emissions / Sustainable transport	CO_2 emission prediction for ICE and hybrid electric vehicles	Linear Regression, Decision Tree, Random Forest, **XGBoost**	ALE (Global), Feature Importance Analysis (Global)	MSE, RMSE, R^2
Include	**132.Zhang et al. (2025)**	Urban sustainability / Regional carbon emissions / Spatiotemporal analysis	Fine-scale energy-related carbon emission estimation and spatiotemporal reconstruction	Linear Regression, SVR, Random Forest Regression, **XGBoost**	SHAP (Global + Local + Interaction)	R^2, RMSE, MAE, PRD
Include	**133.Zhao and He (2025)**	Transportation / Urban transportation systems / Carbon emissions	City-level transportation carbon emission prediction	**CatBoost**, LightGBM, XGBoost, RF, GBR, MLP, ELM, Bayesian Regression, KNN, DT	SHAP (Global + Local + Interaction)	R^2, RMSE, MAE, MAPE
Exclude	134.Yun et al. (2025)	Wastewater treatment / Urban WWTP optimization	Effluent TN concentration and energy-consumption prediction/optimization	MLR, SVM, RF, LightGBM, CatBoost, XGBoost	SHAP, PDP / 2D-PDP	R^2, RMSE
Exclude	135.Zhang et al. (2025)	Transportation / Urban traffic networks / CO_2 concentrations	High-resolution daytime on-road CO_2 concentration and CO_2 increment (ΔCO_2) prediction	RF, ET, LightGBM, XGBoost, AVE, **STK**	SHAP (Global + Local), PDP (Global)	R^2, MAE, RMSE, SMAPE

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Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
Exclude	136.Goto et al. (2025)	Transportation / Plug-in hybrid electric vehicles	Electric mileage ratio (EMR) prediction and factor decomposition	MLR, RF Regression, Gradient Boosting Regression, EBM	EBM (intrinsic XAI / glass-box)	RMSE, MAE
Include	**137.Roy et al. (2025)**	Transportation / Urban traffic / Signalised intersections	Joint prediction of traffic CO_2 emissions and total travel delay at signalised intersections	SVR, RF, XGBoost, GB, MTCNN, **MT2CN**	SHAP (Global + Interaction)	R^2, MAPE, MAE, RMSE
Exclude	138.Ciobotaru et al. (2025)	Predictive maintenance / Air compressor condition monitoring	Air-compressor component condition and fault classification	DNN + SVM, DNN, SVM	SHAP, LIME, PDP	Accuracy, Precision, Recall, F1-score
Exclude	139.Miao et al. (2025)	Sustainable concrete / Recycled aggregate self-compacting concrete	Compressive-strength prediction	M5P, SVR, MLP, RF, XGBoost	SHAP	R^2, RMSE, MAE, MAPE, OBJ
Exclude	140.Ukwuoma et al. (2025)	Renewable energy / Biomass–plastic co-gasification	Hydrogen (H_2) production prediction	agMLP, LARS, Elastic Net, LR, Lasso, Ridge, Huber, RF, XGBoost, DT, ensemble models	SHAP, LIME	MAE, MSE, RMSE, R^2, MSLE, RMSLE
Exclude	141.Escribano et al. (2025)	Hydrogen production / Dry reforming of biogas	Operational-state classification: Activation, Reaction, Irregularity	KNN, QDA, SVM, RF	SHAP, PDP, ICE	Accuracy, F1-score
Exclude	142.Paramasivam et al. (2025)	Renewable energy / Biomass gasification	H_2, CO, LHV and CGE prediction	XGBoost	SHAP	R^2, MSE, MAE
Exclude	143.Yousef et al. (2025)	Renewable energy / Wind energy forecasting	Long-term wind-speed forecasting	LSTM, GRU, LSTM-GRU, BiLSTM-BiGRU + SOA optimization	LIME	MSE, RMSE, MAE, R^2
Exclude	144.Kiouvrekis et al. (2025)	Maritime transport / Vessel performance	Shaft-power estimation/prediction	k-NN, SVM, DT, RF, NN, XGBoost, LightGBM	SHAP	R^2, RMSE, SD, 95% CI
Exclude	145.AlShammari (2025)	Renewable energy / Solar-powered absorption cooling	Coefficient of Performance (COP) prediction and optimization	ANN, ANN-CSO, ANN-MFOA, ANN-WOA	SHAP	RMSE, MAE, MAPE, R^2
Exclude	146.Zhang et al. (2025)	Sustainable pavement / Climate-resilient epoxy asphalt	High-temperature stability and low-temperature crack-resistance prediction	XGBoost, PSO-XGBoost	Feature Importance, ALE	MAE, MAPE, RMSE
Exclude	147.Tieng et al. (2025)	Manufacturing / Facility energy management / Chiller systems	Facility efficiency classification and energy-consumption prediction for energy saving and carbon reduction	Bayesian, SVM, KNN, RF, XGBoost, LSTM, GA	LIME	Accuracy; energy-saving rate
Exclude	148.Zhao et al. (2025)	Multi-energy supply stations / Urban energy systems	Hourly gross transaction value (GTV) / revenue forecasting	DTR, **RF**, SVR, MLP	SHAP	MAE, MSE, RMSE, MAPE, R^2, EV
Not retrieved	149.Wenjie et al. (2025)	Not retrieved	Not retrieved	Not retrieved	Not retrieved	Not retrieved
Exclude	150.Li et al. (2025)	Transportation / Hybrid electric vehicles / Eco-driving	HEV engine operating-state (ICE on/off) classification	LR, NB, SVM, RF, XGBoost	LIME	Accuracy, ROC/AUC
Include	**151.Zheng and Shuang (2025)**	Climate / Country-level carbon emissions	China's carbon-emission forecasting, carbon peaking and	SVM, ANN, RF, XGBoost,	SHAP (Global + Local)	MAE, RMSE, MAPE, R^2

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Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
			neutrality assessment under climate-extreme scenarios	**CatBoost**, BP, LSTM		
Include	**152.Xu et al. (2025)**	Urban planning / Residential land carbon emissions	Residential-land carbon-emission prediction and built-environment driver analysis	**XGBoost**, GBDT, RF, AdaBoost, Logistic	SHAP (Global + Local + Interaction)	R^2, MAE, MSE, RMSE, MAD, MAPE, EVS, MSLE
Include	**153.Han and Lin (2025)**	Environmental economics / Global country-level CO_2 emissions	Long-run CO_2-emission forecasting and EKC analysis	LR, RF, XGBoost, SVR, **ANN**	SHAP (Global + Local)	MAE, MSE, RMSE, R^2
Exclude	154.Tudor et al. (2025)	Energy transition / EU-27 fossil-fuel dependence	Share of fossil fuels (SFF) forecasting through 2030	RF, SVM, XGBoost, Ensemble	SHAP, PDP, interaction analysis	RMSE
Include	**155.Wang et al. (2025)**	Urban sustainability / Digital transformation	Urban carbon-emission-intensity (CEI) prediction and nonlinear driver analysis	OLS, Ridge, LASSO, GBDT, RF, XGBoost, **LGBM**	SHAP (Global, Local, Interaction)	R^2, RMSE, MAE
Include	**156.Chen et al. (2025)**	Regional carbon emissions / China / Carbon-reduction pathways	Provincial carbon-emission prediction and scenario-based emission-path analysis	ARIMA, GM, XGBoost, BRB, DEA-EBRB, Bi-EBRB, **CBRB-IC**	SHAP (Global + Local); intrinsic rule-based interpretability	MAE, RMSE, MAPE, R
Include	**157.Jeaunita et al. (2025)**	Precision agriculture / Smart farming	Carbon-emission prediction + anomaly detection	**FNN** (carbon prediction), Deep Autoencoder (anomaly detection), VGG16 feature extraction	SHAP (Global), LIME (Local)	R^2, MSE, RMSE; Accuracy, Precision, Recall, F1
Include	**158.Liang et al. (2025)**	Carbon market / Emissions trading	Carbon-price forecasting	LR, LGBM, ANN, KNN, LSTM, ARIMA, EEMD–GAMI_NET–LSTM–ARIMA	ALE (Global)	RMSE, MAE, R, DA
Exclude	159.Olawumi & Oladapo (2025)	Sustainable energy management / Net Zero	Energy-demand/consumption prediction and emission-reduced energy allocation	GBM, RF, ANN-based framework	SHAP, LIME	MAE, RMSE, R^2, Precision, Recall
Exclude	160.Mungan & Arpa (2025)	Transportation / Vehicle emissions	Vehicle CO_2-emission prediction	LR, Lasso, KNN, RF, CatBoostRegressor	-	MSE, RMSE, MAE, R^2
Exclude	161.Elbasi et al. (2025)	Smart agriculture / Green AI	Energy-efficient crop-yield forecasting and resource management	Lightweight AI, compressed models, FL, transfer learning, edge/fog and hybrid Green AI approaches	-	-
Exclude	162.Inqiad et al. (2025)	Sustainable construction / Soilcrete materials	Density, shrinkage and strain prediction of soilcrete	GEP, XGB	SHAP, ICE	R^2, MAE, RMSE, a20-index, PI, OF
Exclude	163.Das et al. (2025)	Solar thermal energy / Parabolic trough solar collector (PTC)	PTC outlet-temperature prediction	M5P Model Tree, Elastic.Net Regression	Intrinsic interpretability (M5P, Elastic.Net)	R^2, MAE, RMSE
Exclude	164.Fuqaha et al. (2025)	Sustainable construction / Concrete with non-potable water	Concrete compressive-strength prediction	DT, GP-SE, GPR, SVM, Bagged Trees, Boosted Trees, RF, LSBoost	MDI, PDP	R^2, MSE, RMSE, MAE, MAPE, NRMSE

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Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
Exclude	165.Bhambare et al. (2025)	Mining / Sustainable environmental management	Review of AI applications for environmental monitoring, energy optimization and emission reduction	-	-	-
Exclude	166.Ersöz et al. (2025)	Drone technology / Li-ion battery management	Li-ion battery performance prediction during drone flights	CNN, RNN, CNN+RNN, Ensemble	MAE, RMSE, R^2	PDP
Include	**167.Guo et al. (2024)**	Transportation / Vehicle emissions	Vehicle CO_2-emission prediction in Canada	KNN, SVR, **GBM**, DT, RF, LASSO	SHAP (Global), ALE (Global)	R^2, RMSE, rRMSE, MAE, MAPE
Exclude	168.Asim et al. (2024)	Smart electricity markets	Review of price, demand, load, carbon-emission and supply forecasting	-	-	-
Exclude	169.Bao et al. (2024)	CO_2 utilization / Methanation process optimization	CO_2 conversion ratio, CH_4 selectivity and exergy-efficiency prediction	SVR, DNN, XGBoost	SHAP, PDP	R^2, RMSE
Exclude	170. Mahamoodally et al. (2024)	Smart buildings / Occupancy estimation	Imbalanced occupancy classification with domain adaptation	SER, STRUT, SER, STRUT, GOSDT-DA, SER-STRUT, IMB-SER	SHAP, interpretable IF–ELSE rules	Accuracy, Precision, F1-score, ROC-AUC
Include	**171.Sayed et al. (2024)**	Carbon market / Carbon trading	Carbon-price forecasting	MLSO-LSTM, LSTM, RF, Gradient Boosting + swarm-optimized LSTM variants	SHAP	RMSE, R^2, MAE, MAPE, MSE
Exclude	172.Zhang et al. (2024)	Electric vehicles / Urban transportation	Review of EV energy-consumption prediction and modelling	-	-	-
Exclude	173.Liu et al. (2024)	Air-quality monitoring / Energy conservation	Air-quality status classification and CEP rule extraction	DT, LightGBM, Enhanced DT, Process Mining, Petri Net	Intrinsic DT rules / interpretable rule extraction	Accuracy, Precision, Recall, F1-score, AUC
Exclude	174.Zhang et al. (2024)	Urban transportation / Street environment	On-road CO_2 concentration prediction and spatial mapping	RF, XGBoost, LGB, CatBoost, EL Fusion	SHAP (LGB-SHAP)	R^2, MAE, MAPE
Exclude	175.Guo et al. (2024)	Sustainable construction / Ultra-high-performance geopolymer	Compressive-strength prediction + multi-objective low-carbon material design	LR, Ridge, SVR, KNN, RF, XGBoost, LightGBM, CatBoost	Knowledge graph-guided interpretability	MAE, RMSE, MAPE, R^2
Exclude	176.Shokrollahi et al. (2024)	Carbon capture and storage / Subsurface CO_2 storage	CO_2 solubility prediction in brine solutions	RF	SHAP, SHAP interaction values, Friedman H-statistics	AARD, RMSE, R^2
Exclude	177.Ngeni et al. (2024)	Urban transportation / Bike-sharing systems	Bike-sharing station demand/capacity prediction	Random Forest	SHAP	MSE, MAE
Exclude	178.Elmousalami et al. (2024)	Construction / Irrigation canal rehabilitation / Green AI	Project cost, cost variance, duration and duration-variance prediction	ANN, MLR, PR, SVM, DT, ET, LGBM, GB, XGB, BDT, RF, AdaBoost	SHAP	MAPE, RMSE, MSE, R^2, adjusted R^2
Exclude	179.Arbaoui et al. (2024)	Energy storage / Lithium-ion batteries	Battery State of Health (SOH) forecasting and abnormal degradation detection	E-LSTM, CNN–LSTM	SHAP	MAE, MSE, MAPE
Include	**180.Khan et al. (2024)**	Green buildings / Sustainable construction /	Energy consumption, CO_2-emission, and thermal-discomfort prediction	LGBM, **BO-LGBM**, BO-XGB	LIME (Local)	R^2, MAPE

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Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
		Building energy performance				
Exclude	181.Alyami et al. (2024)	Sustainable construction / RHA concrete	Compressive-strength prediction of rice-husk-ash concrete	Ridge Regression, RF, XGBoost, LightGBM	SHAP	R, MAE, RMSE, RRMSE, ρ, a20-index, OF
Exclude	182.Nallakarup pan et al. (2024)	Solar energy / Power-system management	Solar AC power-output prediction	LR, RF, Gradient Boost, Extra Gradient Boost	LIME, PDP	R^2, Variance, RMSE
Include	**183.Deng et al. (2024)**	Carbon market / Emissions trading / Hubei carbon market	Multi-step carbon price trend forecasting (upward/downward; 1-, 5-, and 10-step ahead)	MARS, SVM, RF, ELM, XGBoost, LSTM, LightGBM, CEEMDAN-LightGBM-S, CEEMDAN-LightGBM, CEEMDAN-TFS-LightGBM, CEEMDAN-BOA-LightGBM, **CEEMDAN-TFS-BOA-LightGBM**	SHAP (Global + Local)	AUC, Accuracy, Precision, Recall, F1-score
Exclude	184.Wang et al. (2024)	Wastewater treatment / Intelligent aeration control	Aeration/air-demand prediction for low-carbon WWTP operation	LIN, KNN, SVM, RF, LGBM, DNN, RNN, LSTM + multimodal variants; RF-M	SHAP	MAPE, RMSE, R^2
Include	**185.Luo et al. (2024)**	Land use / Regional sustainability / Carbon emissions	Multi-scale land-use-based carbon-emission prediction	LR, MLP, KNN, SVM, ETR, RF, DT, XGBoost, **Optimized ETR**	SHAP (Global + Local)	R^2, MSE, RMSE, MRE, MAE, MBE
Exclude	186.Lee et al. (2024)	Maritime transportation / Smart ships	Fuel-consumption prediction and derived CO_2-emission estimation	RFR, LightGBM, LSTM	SHAP	R^2, MAPE, RMSE
Include	**187.Mangla et al. (2024)**	Energy economics / Energy risk management / Global GHG emissions	Total GHG-emission modelling and identification of emission drivers	RF, MLP, DNN	SHAP (Global + Local)	R^2, Adjusted R^2 (regression validation); Support, Confidence, Lift (association rules)
Exclude	188.Lee (2024)	Solar energy / Organic photovoltaics	Fill-factor prediction of non-fullerene organic solar cells	GBDT	SHAP	R^2, RMSE, MAE
Exclude	189.Sison et al. (2024)	Biomass gasification / Chemical looping gasification	H_2 yield and char-yield prediction	SVM, RF, GBR	SHAP, PDP	R^2, RMSE, MAE
Exclude	190.Vellayappan et al. (2023)	Dry methane reforming / Heterogeneous catalysis	Syngas ratio, CH_4 conversion and CO_2 conversion prediction	RF, GBR, CatBoost	SHAP, PDP, permutation/Gini importance	R^2, RMSE
Include	**191.Yao et al. (2023)**	Urban carbon emissions / Urban sustainability	CO_2 emission turning-point forecasting and identification of socioeconomic drivers	EKC, ARIMA, ECM, DT, **GA-EKC**, SR, XGBoost	SHAP (Global + Local)	R^2, Adjusted R^2
Include	**192.Assael et al. (2023)**	Corporate sustainability / Greenhouse gas emissions / Carbon accounting	Estimation of non-reported corporate Scope 1 and Scope 2 GHG emissions	**LightGBM**	SHAP (Global + Local)	R^2, RMSE, MAE
Include	**193.Zhang et al. (2023)**	Buildings / Building energy efficiency / GHG emissions	Residential building energy consumption and GHG emission prediction	**LightGBM**, XGBoost, RF, SVR	SHAP (Global + Local), PDP (Global)	R^2, MSE, RMSE, MAE
Exclude	194.Fatahi et al. (2023)	Cement industry / Rotary-kiln process control	Kiln feed-rate and ID-fan-current prediction	XGBoost, RF, SVR	SHAP	R^2, RMSE

A. Records retrieved from Scopus (n = 203) (including records also indexed in WoS)						
Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
Exclude	195.Shah et al. (2022)	Sustainable construction / Alkali-activated materials	28-day compressive-strength prediction	Ridge Regression, RF, LightGBM, XGBoost	SHAP	R^2, MAE, RMSE
Exclude	196.Heo et al. (2022)	Petrochemical industry / Offshore renewable energy / Net-zero planning	Offshore wind-power forecasting and downstream net-zero carbon assessment	VAE	t-SNE, DBSCAN	VAE total loss; latent-cluster correlations
Exclude	197.Moraliyage et al. (2022)	Building energy efficiency / Energy Conservation Measures / M&V	Building energy-consumption prediction and energy-savings estimation	XGBoost	SHAP	CV(RMSE), R^2, MAPE, NMBE
Exclude	198.Wenninger et al. (2022)	Building energy / Residential energy performance	Annual building Final Energy Performance (FEP) prediction	QLattice, XGB, ANN, SVR, MLR	QLattice intrinsic interpretability, Permutation Feature Importance	CV, RMSE, MAE, MAPE, MSE
Exclude	199.González-González et al. (2022)	Industrial carbon footprint / Banking / NLP	Bank-transaction activity classification followed by carbon-footprint estimation	SVC, RF, LSTM	LIME	Accuracy, Precision, Recall
Exclude	200.Mouakher et al. (2022)	Smart energy / Residential energy consumption	Daily, weekly and monthly energy-consumption forecasting	LR, RF, EXPECT (LSTM)	PDP	MSE, MAE
Exclude	201.Takalo-Mattila et al. (2022)	Steel manufacturing / Quality control	Steel surface-defect classification	LightGBM (GBDT)	SHAP	AUC-ROC, MCC
Exclude	202.Labe and Barnes (2021)	Climate science / Climate-signal attribution	Year/decade prediction from near-surface temperature maps under different climate forcings	ANN	LRP	R^2, correlation
Include	**203.Crespo et al. (2021)**	Regional/sectoral carbon emissions / Environmental policy	Total and sector-specific EU28 CO_2-emission prediction	NN/BP, NN/BP-BFE, NN/BP-RReliefF/BFE, **Proposed NN/BP-RReliefF framework**, GBM-BFE, RF-BFE, SVM-BFE	LIME (Local), RReliefF (Global Feature Importance	RMSE, MAPE
Scopus subtotal: 203 records; 69 included, 125 excluded, 9 full texts not retrieved.						
B. Records retrieved only from Web of Science (n = 9) (Numbers refer to the original WoS search results; records duplicated in Scopus were removed.)						
Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
Not retrieved	12.Kim et al. (2026)	Not retrieved	Not retrieved	Not retrieved	Not retrieved	Not retrieved
Exclude	13. Hou et al. (2026)	Urban climate / Built environment	Seasonal block-scale land-surface-temperature prediction	LR, Ridge, Lasso, DT, RF, Extra Trees, KNN, XGBoost, LightGBM	SHAP	R^2, RMSE, MAE
Exclude	16.Do et al. (2026)	Solar energy systems / Renewable energy	Review of ML applications in solar forecasting, optimization, fault detection and energy management	Review	Review	Review
Exclude	20.Tahera et al. (2026)	Sustainable concrete / Circular construction	Mechanical and durability performance prediction with LCA-based embodied-carbon assessment	RF, XGBoost, LightGBM, ANN, SNN, LR	SHAP	R^2, RMSE, MAE

A. Records retrieved from Scopus (n = 203) (including records also indexed in WoS)						
Decision	**Author(s)**	**Application area**	**Prediction task**	**Model(s)**	**XAI method(s)**	**Performance metrics**
Exclude	33.Şişman & Yavuz (2026)	Sustainable transportation / Electric vehicles	EV adoption-intention classification	LR, DT, RF, SVM, XGBoost	SHAP	Accuracy, Precision, Recall, F1-score, ROC-AUC
Exclude	53.Akdoğan (2025)	Corporate carbon disclosure / Sustainable finance	Prediction of voluntary carbon-disclosure behavior	KNN, NB, SVM, DT, RF, XGBoost	XGBoost feature importance	Accuracy, Precision, Recall, F1-score, ROC-AUC
Include	**70.Liu et al. (2025)**	Urban transportation / Metro–bus combined travel	Trip-level CO_2 emission reduction rate prediction	GBDT, XGBoost, LightGBM, **CatBoost**	SHAP (Global + Local), PDP (Global)	R^2, RMSE, MAE
Include	**75.Zhou et al. (2024)**	Office buildings / Operational carbon emissions	Building carbon emission, PV carbon offset, and net carbon-emission prediction	XGBoost, **LightGBM**	SHAP (Global + Local + Interaction)	R^2, MAPE, MAE
Exclude	84.Chen & Jia (2024)	Coal market / Energy transition / Chinese industry	Chinese coal-consumption and coal-price-index prediction	LASSO, MLR, SVM, **RF**, XGBoost, ANN, ELM, LSTM	SHAP, PCMCI	R^2, RMSE, MSE, MAE
WoS-only subtotal: 9 records; 2 included, 6 excluded, 1 full text not retrieved.						
Total: 212 records; 71 included, 131 excluded, 10 full texts not retrieved.						